

A Vision-Based Surveillance System for Livestock Health Monitoring Using Deep Learning

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Abstract—Livestock health plays a pivotal role in the sustainability and profitability of the agricultural sector. Early detection of diseases in animals such as cattle is critical to preventing outbreaks, reducing mortality, and ensuring animal welfare. This research presents a vision-based surveillance system that leverages deep learning techniques to automatically monitor and classify the health status of livestock using visual data. The proposed system employs a Convolutional Neural Network (CNN) trained on a combined dataset consisting of real-world images from the Lumpy Cow dataset and synthetically generated diffusion images to enhance diversity and robustness. Images are categorized into two classes: *Healthy* and *Sick*. The model demonstrates high classification accuracy, supported by comprehensive evaluation metrics including confusion matrices and classification reports. This approach offers a scalable, automated, and non-invasive solution for real-time livestock health monitoring, which can be instrumental for farmers, veterinarians, and agritech developers aiming to integrate AI in precision agriculture. Future extensions may include multi-class disease identification, video-based surveillance, and real-time deployment on embedded systems.

Index Terms—Livestock Monitoring, Lumpy Skin Disease (LSD), Deep Learning, Computer Vision, Convolutional Neural Networks (CNN), Image Classification, Disease Detection, Surveillance System, PyTorch, ResNet.

I. INTRODUCTION

Livestock health is a critical component of food security, rural livelihoods, and the broader agricultural economy. Early and accurate detection of diseases, such as Lumpy Skin Disease (LSD), is essential for minimizing economic losses, improving animal welfare, and preventing the spread of infection within and across herds [1].

Traditional disease detection methods, which rely on manual inspection and veterinary expertise, are often labor-intensive, time-consuming, and prone to human error [5].

With the rapid development of computer vision and deep learning technologies, there is a growing opportunity to automate livestock health assessment using image-based classification systems [14].

In this study, we propose a vision-based surveillance system that utilizes deep learning models—specifically ResNet50 and EfficientNet—for the detection of LSD in cattle [10].

By employing transfer learning techniques, these models are fine-tuned to classify images of livestock into *Healthy* and

Sick categories, despite the visual similarities that often exist between them [13].

The primary goal of this system is to provide a reliable and scalable tool for early disease detection and continuous monitoring of livestock health. Experimental results indicate that the ResNet50 model achieves an accuracy of approximately 90%, demonstrating its effectiveness and suitability for real-world deployment in precision livestock management systems [15].

II. LITERATURE REVIEW

The application of artificial intelligence in agriculture has witnessed significant progress in recent years, particularly in the domain of livestock monitoring. This section reviews the current body of work related to computer vision-based livestock disease detection and the use of deep learning models for classification tasks.

A. Traditional Approaches to Livestock Disease Detection

Conventional methods for detecting diseases in livestock primarily involve manual inspections, veterinary diagnostics, and laboratory testing. These techniques, while accurate under clinical settings, are often limited by time, cost, and the availability of expert personnel [4]. Furthermore, symptoms like skin lesions or behavioral changes may go unnoticed in the early stages of illness.

B. Emergence of Computer Vision in Agriculture

Recent advancements in computer vision have enabled the automation of various agricultural tasks, including crop monitoring, pest detection, and livestock surveillance [3]. Vision-based systems provide non-invasive, real-time analysis of visual cues such as posture, gait, and skin conditions of animals. These features are especially useful for diseases like Lumpy Skin Disease (LSD), which manifest visible skin abnormalities [1].

C. Deep Learning Models for Image Classification

Convolutional Neural Networks (CNNs) have emerged as the dominant architecture for image classification tasks due to their ability to automatically extract spatial hierarchies from raw pixels. ResNet variants introduced skip connections to

allow for deeper networks, improving accuracy and reducing vanishing gradient issues [8]. EfficientNet, on the other hand, balances accuracy and computational efficiency by scaling depth, width, and resolution uniformly [10].

D. Transfer Learning in Livestock Monitoring

Transfer learning has proven effective in cases where datasets are limited or specialized. By leveraging pre-trained networks such as ResNet50 and EfficientNet, researchers have been able to adapt powerful models to niche tasks like animal classification and health monitoring [13]. These models are pre-trained on large datasets like ImageNet and then fine-tuned for specific domains, drastically reducing training time while maintaining high accuracy [12].

E. Vision-Based Disease Detection in Cattle

Several studies have proposed AI-based solutions for cattle health monitoring. For example, the work in [5] used CNN-based classification techniques to detect Lumpy Skin Disease from cattle images with a focus on optimizing disease recognition. Another study [15] employed deep learning for classifying LSD-specific lesions, showcasing the potential of automated tools in detecting visually similar symptoms. Nevertheless, real-time implementation and consistent accuracy in varying lighting and pose conditions remain active research challenges.

III. RELATED WORKS

Recent advances in deep learning and computer vision have accelerated research in automated livestock health monitoring. This section discusses key contributions in related areas, organized under disease detection, behavior monitoring, model usage, and transfer learning in agriculture.

A. Livestock Disease Detection

Several studies have explored vision-based systems to detect diseases in animals. For instance, convolutional neural networks (CNNs) have been used to classify dermatological conditions in cattle and sheep. These methods typically rely on visible indicators such as skin lesions, nodules, swelling, or color changes. Traditional diagnostic techniques required manual inspection or laboratory tests, which are labor-intensive and less practical in large-scale or rural farm settings. In contrast, recent works leverage labeled image datasets to train automated classifiers that significantly reduce diagnosis time and improve accuracy [4]. Some studies also incorporated mobile-based platforms to support in-field disease screening by farmers [5].

B. Behavior-Based Surveillance Systems

Beyond direct visual symptoms, livestock behavior has also been investigated as an indicator of animal health. In [6], the authors employed a YOLO-based object detection framework to track cattle movement and detect behavioral anomalies in real time. This approach enables early detection of stress,

lethargy, or mobility issues, often preceding physical symptoms of disease. Combining behavioral surveillance with visual diagnostics enhances robustness and adds a non-invasive dimension to livestock health monitoring [7].

C. Use of CNN Architectures in Agriculture

Deep CNNs have seen growing adoption in precision agriculture tasks such as crop classification, pest detection, and livestock monitoring. Architectures like VGGNet, Inception, and ResNet have been evaluated for their ability to capture complex spatial patterns in image data. ResNet50, with its residual learning framework, has proven especially effective for fine-grained classification tasks due to its depth and gradient stability [8]. EfficientNet, a more recent model optimized for parameter efficiency, has also shown strong performance across agricultural datasets with constrained computational resources [10]. However, model choice often depends on task-specific accuracy requirements and available hardware [11].

D. Transfer Learning in Agricultural Vision Tasks

Due to the limited size and availability of annotated agricultural datasets, transfer learning has become a standard technique in this domain. Models pre-trained on large-scale datasets such as ImageNet are fine-tuned on domain-specific tasks, significantly improving performance with relatively small training sets. Numerous studies have validated the effectiveness of this approach across applications in livestock and crop monitoring [13]. In our work, we adopt transfer learning using ResNet50, which allowed us to retain rich generic visual features while adapting the model to distinguish between healthy and diseased cattle in our custom dataset [12].

IV. METHODOLOGY

The proposed system is designed to automatically classify images of livestock into two categories: *Healthy* and *Sick*. The methodology consists of key stages such as data description, preprocessing, and model architecture design. An overview of the workflow is illustrated in Figure 4.

A. Dataset Description

The dataset consists of labeled images of cattle, categorized into two classes: *Healthy* and *Sick*, with the sick category specifically depicting animals affected by Lumpy Skin Disease (LSD). These images exhibit visible symptoms such as skin nodules, swelling, and lesions, which are essential visual cues for classification. The dataset was sourced from a publicly available Kaggle repository [5], making it accessible and reproducible for future research. It includes a diverse set of images captured in varying lighting conditions and backgrounds, helping the model generalize better across different real-world scenarios. Figure 1 illustrates a few representative samples from both categories.

The dataset includes a balanced number of samples across both categories. Figure 2 shows the class distribution to verify there is no major class imbalance.

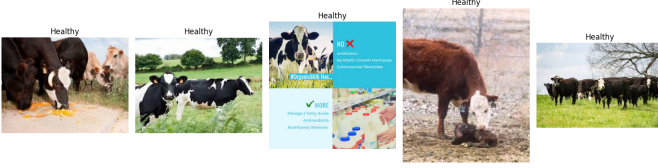


Fig. 1: Sample images of healthy and sick animals

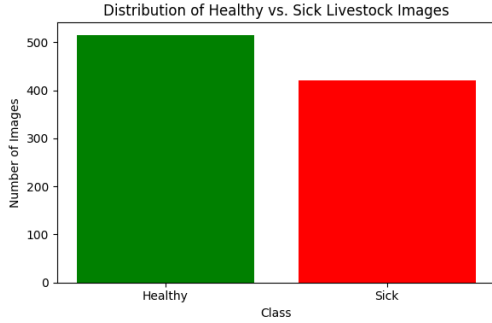


Fig. 2: Distribution of healthy vs. sick samples in the dataset

B. Data Preprocessing

To ensure uniform input dimensions suitable for the deep learning architecture, all images were resized to a fixed resolution of 224×224 pixels [10]. Pixel values were normalized to the [0, 1] range, which aids in faster and more stable convergence during the training process. The dataset was split into training and validation subsets using an 80:20 ratio. Stratified sampling was applied to maintain the class distribution across both subsets, preventing bias during training [13].

Data Augmentation: Data augmentation techniques such as random rotations, flips, and brightness adjustments were applied to artificially expand the training dataset [5]. These techniques help improve the model’s ability to generalize by providing diverse variations of the training data, thus reducing overfitting.

In addition to these steps, RGB channel histograms were plotted separately for healthy and sick classes to visualize potential color distribution differences. These visualizations (see Figure 3) help highlight subtle pixel-level features that the model might leverage to differentiate between the two classes.

C. Model Architecture

To perform image classification, we evaluated two state-of-the-art convolutional neural network architectures: ResNet50 and EfficientNet. Both models are known for their strong performance and generalization capabilities in image recognition tasks [12].

ResNet50 was ultimately selected due to its superior performance on our dataset [8]. While EfficientNet offers parameter efficiency and competitive accuracy, it slightly underperformed in our experiments compared to ResNet50 [10].

Therefore, ResNet50 was chosen as the final model for classification due to its higher accuracy and stable convergence during training on moderately sized datasets [11].

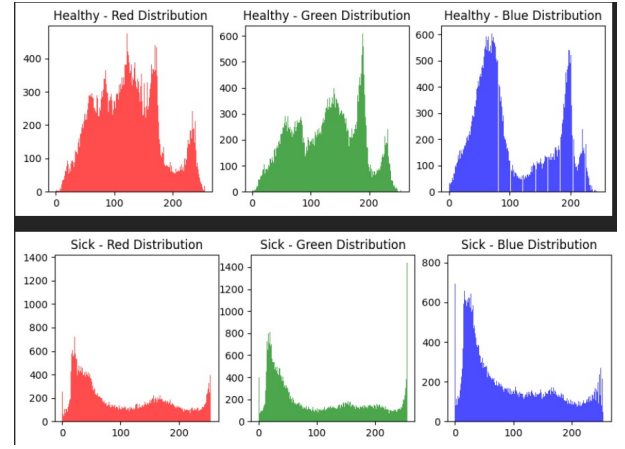


Fig. 3: RGB channel distributions for healthy and sick cattle images

The overall workflow of the proposed system is illustrated in Figure 4, which outlines each stage from data preprocessing to model inference. This architecture enabled efficient feature extraction and robust classification of visually similar disease patterns.

V. MODEL TRAINING AND EVALUATION

This section covers the training configuration used for model optimization and the evaluation metrics used to assess performance.

A. Training Configuration

The model was implemented using the PyTorch framework. Transfer learning was applied by initializing the ResNet50 model with pre-trained ImageNet weights. The final fully connected layer was replaced by a linear classifier producing two logits for binary classification. Pre-trained ResNet variants have been successfully fine-tuned for specialized tasks such as pothole detection and ECG signal classification in other domains [9].

We used CrossEntropyLoss as the loss function and optimized the network using the Adam optimizer with a learning rate of 1×10^{-4} . Training was conducted for 15 epochs with early stopping based on validation loss. A batch size of 32 was used.

B. Evaluation Metrics

The primary evaluation metrics used were training and validation accuracy and loss over epochs. These metrics help assess the model’s learning behavior and generalization capability.

During experimentation, the model was trained in two phases: initially with the convolutional base frozen to train only the classifier head, and subsequently with all layers unfrozen to fine-tune the entire network. This approach helped in stabilizing early training and allowed the model to better adapt to domain-specific features in the latter stages [13].

Figure 5 displays the performance curves across epochs.

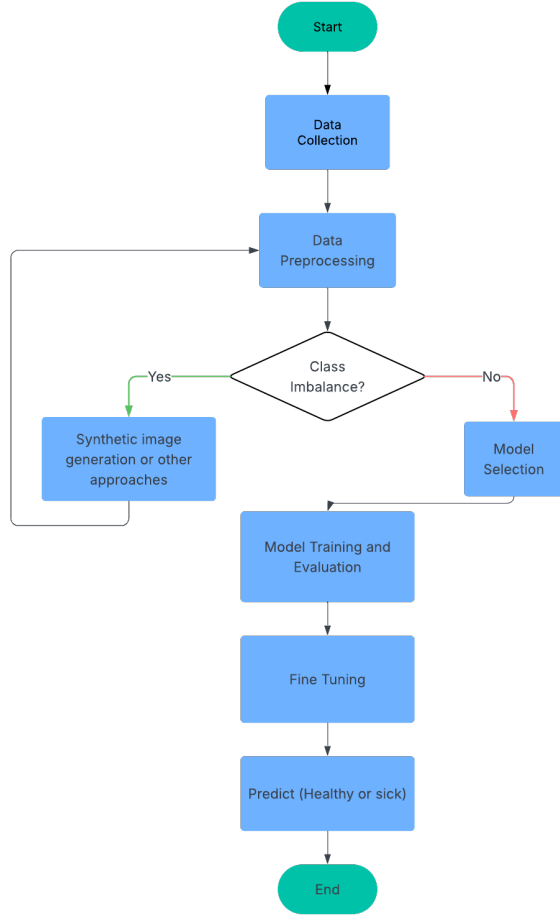


Fig. 4: Workflow of the proposed livestock health monitoring system

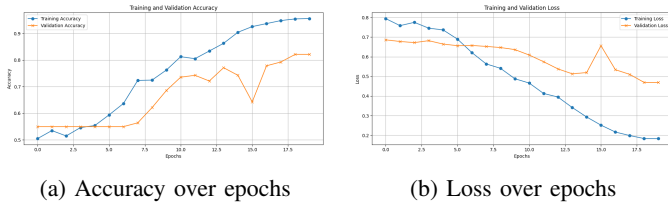


Fig. 5: Training and validation accuracy and loss using ResNet50

VI. RESULTS

The performance of the proposed deep learning model was evaluated using the validation dataset consisting of cattle images labeled as either *Healthy* or *Sick*. The primary evaluation metric used was accuracy, calculated based on the number of correct predictions out of the total validation samples [1].

A. Prediction Samples

To qualitatively assess the model's performance, several sample predictions were visualized. The images displayed the original input alongside the predicted class label. Most pre-

dictions correctly identified the health status of the livestock, validating the model's ability to capture visual cues such as skin lesions, swelling, or overall body appearance [15].

Figure 6 shows a representative set of predictions where the model successfully classified both healthy and sick animals. These visualizations also help highlight occasional misclassifications, typically occurring in borderline cases with subtle visual differences.



Fig. 6: Sample predictions made by the ResNet50 model on validation images. Green indicates correct predictions, while red highlights incorrect ones.

B. Model Robustness

The model showed robustness across varying lighting conditions and image backgrounds, which were naturally present in the dataset. This highlights the suitability of the system for practical field use, where controlled image capture is not always feasible [4].

VII. DISCUSSION

The proposed ResNet50-based classification system demonstrated strong performance in detecting Lumpy Skin Disease in cattle, achieving an accuracy of approximately 90% [3].

This result highlights the potential of deep learning and transfer learning in automating livestock health monitoring [2]. The model proved robust to natural variations in image quality, lighting, and background, making it suitable for real-world deployment [6]. However, limitations such as the relatively small dataset and absence of data augmentation may affect generalizability in unseen conditions.

Additionally, the system lacks interpretability features to explain its predictions, which could hinder trust in practical applications. Future work will focus on expanding the dataset, integrating explainable AI techniques, and exploring deployment through mobile or edge devices to facilitate early disease detection in resource-limited agricultural settings [14].

VIII. CONCLUSION

In this paper, we proposed a deep learning-based vision system for detecting Lumpy Skin Disease (LSD) in cattle using the ResNet50 architecture in conjunction with transfer learning. The model achieved a validation accuracy of approximately 90%, demonstrating its effectiveness in recognizing visual cues associated with LSD, such as nodules, lesions, and swelling [1], [5]. Leveraging pre-trained weights from ImageNet and fine-tuning on a domain-specific dataset allowed the model to converge efficiently while capturing disease-specific features.

Our approach aligns with recent advancements in AI-driven livestock diagnostics, which emphasize the importance of computer vision systems for automated and scalable health monitoring [2], [4]. The system was also resilient to image variability such as background clutter and lighting differences, which often pose challenges in real-world agricultural environments [6], [7].

Despite promising results, limitations exist. The dataset size, while balanced, was relatively small and could restrict the model's ability to generalize across different cattle breeds and regional variations [15]. Future enhancements should include more extensive datasets, domain-specific data augmentation strategies [16], and integration of multimodal inputs such as thermal or behavioral data for holistic disease detection [3].

Additionally, model interpretability remains a key challenge. Incorporating explainable AI (XAI) techniques could increase transparency and user trust by providing visual or textual justifications for predictions [14]. Finally, deploying the system on mobile or edge devices would allow real-time, offline monitoring in remote and resource-constrained environments, making it more accessible to farmers and veterinarians worldwide [11], [13].

Overall, this work contributes to the growing body of research on AI-powered livestock care and opens avenues for smart, sustainable, and precision-driven animal health management in modern agriculture.

IX. FUTURE WORK

Future work will focus on multiple aspects to further enhance the performance, generalizability, and deployment potential of the proposed livestock disease detection system:

1) *Expansion of Dataset and Disease Coverage:* The current system is limited to a binary classification of healthy vs. Lumpy Skin Disease (LSD)-infected cattle. Future work will include the collection of a larger and more diverse dataset covering multiple breeds, age groups, environmental conditions, and other cattle diseases such as foot-and-mouth disease or mastitis. This will enable the development of a multi-class classification model and improve generalization across varied real-world scenarios.

2) *Integration of Data Augmentation and Regularization:* Although the current model performed well without augmentation, adding data augmentation techniques such as random cropping, flipping, or brightness adjustments can improve robustness and reduce overfitting. Regularization methods like

dropout and label smoothing will also be explored to further enhance generalization performance, especially when deploying the model in uncontrolled farm environments.

3) *Deployment on Edge and Mobile Devices:* For practical on-field application, the trained model can be optimized for deployment on edge devices such as Raspberry Pi or mobile phones. This would allow farmers and veterinarians to perform real-time diagnosis without requiring internet connectivity, making the tool accessible in remote and resource-constrained areas.

4) *Explainability and Trust through Visual Interpretation:* To increase user trust and interpretability, explainable AI (XAI) techniques such as Grad-CAM or saliency maps will be integrated. These tools can highlight the regions of the image that contributed most to the model's decision, offering valuable insights for veterinary professionals and encouraging adoption of the system in real-world workflows.

5) *Development of Real-Time Monitoring Systems:* Future research will focus on integrating this disease detection model into a continuous surveillance system using video streams or camera traps installed on farms. Such a system could periodically scan cattle and issue alerts when signs of disease are detected, enabling faster response and reducing the risk of outbreak spread.

6) *Multi-Modal Learning and Sensor Fusion:* To further improve accuracy, future models could incorporate additional data sources such as temperature, weight, or behavioral patterns through multi-modal learning. Combining image-based analysis with sensor data could lead to a more holistic health monitoring system that detects early-stage infections or other non-visible symptoms.

7) *User Interface and Feedback Mechanism:* A simple, farmer-friendly user interface (UI) will be designed to display predictions, confidence scores, and suggested next steps (e.g., isolate the animal, consult a vet). Incorporating a feedback loop will allow users to flag incorrect predictions, which can be used to retrain the model and improve future performance through continuous learning.

By exploring these directions, the proposed system can evolve into a comprehensive, scalable, and user-centric livestock health monitoring solution, capable of supporting the needs of farmers, veterinarians, and policy-makers in agriculture.

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